



MANE 2720 Fluid Mechanics

Dr. Ozgur Tumuklu

Fall 2025

1 Course Information

Course Title Fluid Mechanics
Course Number MANE 2720
Credit Hours 3
Semester/ Year Fall 2025
Meeting Days Monday + Thursday: 16:00 - 17:20 PM

Room Location Darrin, Room 324

Course Website <https://lms.rpi.edu>

Announcements, supplementary course materials, and discussion will be posted on this website.

Prerequisites ENGR 1100 Introduction to Engineering Analysis and PHYS 1100 Physics I.

Corequisite MATH 2400 Introduction to Differential Equations.

Advanced Standing Engineering.

Instructor Prof. Ozgur Tumuklu

Office: Jonnson Engineering Center (JEC) 2044

Email: tumuko@rpi.edu

Phone: 518.276.7547

Web: <https://mane.rpi.edu/people/faculty/ozgur-tumuklu>

Office Hours:¹

Monday (**Hybrid**): 9:00 - 10:00 AM

Webex Meeting link: <https://rensselaer.webex.com/meet/tumuko>

Thursday (**In person**): 12:30 - 1:30 PM

Or by appointment if it is urgent.

Teaching Assistant Shorob Alam Bhuiyan

Office: Online

Webex Meeting link: <https://rensselaer.webex.com/meet/bhuiys2>

Email: bhuiys2@rpi.edu

Office Hours:

Friday: 4:00 - 6:00 PM

By Appointment

¹Office hours are subject to change. Changes will be announced in class and LMS.



Teaching Assistant Charles Carrillo

Office:

CII 4225

Email:

carric3@rpi.edu

Office Hours:

Wednesday:

1:00 PM - 3:00 PM

By Appointment

Textbook *Fox & McDonald's Introduction to Fluid Mechanics*
Philip J. Pritchard and John W. Mitchell
10th. ed.,² ISBN: 978-1-119-60376-4

Important Dates

Sep. 1: Labor Day-no classes

Sep. 5 FRIDAY: We have class at 4:00 PM at Darrin 324.

Sep. 12: The last day for all students to drop a course without being issued a W grade, or without penalty. Deadline for completion of NE/I grade expectations related to Spring 2024 courses.

Oct. 6: First Exam.

Oct. 13: Columbus Day-no classes

Nov. 6: Second Exam.

Nov. 7: The last day for undergraduate and graduate students to drop a course with a W grade.

Nov. 24-28: Thanksgiving recess

Dec. 11: Third Exam.

Dec. 16-22: Final Exams

[Additional Registrar Dates and Deadlines \(linked\)](#)

2 Course Outline

Course Description

Explore the fundamental aspects of fluid mechanics, encompassing fluid properties, statics, dynamics of both incompressible viscous and inviscid flows, control volume formulations of continuity, momentum, and energy equations, dimensional analysis, viscous flow in pipes, boundary layers, drag, momentum integral theory, and compressible flows.

Student Learning Outcomes

Students who successfully complete this course will have the ability to demonstrate: (i) proficiency in identifying, formulating, and solving engineering problems, (ii) a solid understanding of fluid mechanics, and (iii) a comprehensive knowledge of flow physics.

For RPI policy for Institutional Research and Assessment, can be found in this [link](#).

Outline

Chapter 1: Introduction

Chapter 2: Fundamental Concepts

²10th edition is preferable but older editions will also be suitable. Digital versions are permitted.



Chapter 3:	Fluid Statics
Chapter 4:	Basic Equations in Integral Form for a Control Volume
Chapter 5:	Introduction to Differential Analysis of Fluid Motion
Chapter 6:	Incompressible Inviscid Flow
Chapter 7:	Dimensional Analysis and Similitude
Chapter 8:	Internal Incompressible Viscous Flow
Chapter 9:	External Incompressible Viscous Flow

3 Course Calendar

The following table shows a tentative plan for the course. It may be subject to *slight* changes during the semester.

Week #	Reading Assignments from Textbook	Homework Assignment
1	Chapters 1.1–1.6	1
2	Chapters 2.1–2.7	2
3	Chapter 3.1–3.4	-
4	Chapter 3.4–3.7	3
5	Chapter 4.1–4.3	-
6	Chapter 4.4–4.9	4
Midterm Exam 1 : October 6, 2025 on Chapters 1–3 and part of Chapter 4		
7	Chapter 5.1–5.2	-
8	Chapter 5.3–5.5	5
9	Chapter 6.1–6.7	-
10	Chapter 7.1–7.5	6
Midterm Exam 2 : November 6, 2025 on Chapters 4–6		
11	Chapter 8.1–8.5	–
12	Chapter 8.6–8.8	7
13	Chapter 9.1–9.3	–
14	Chapter 9.4–9.7	8
Midterm Exam 3 : December 11, 2025 on Chapters 7–9		
Final Exam: December 22, 2025 Time:3:00-6:00 PM Location: SAGE 3303 on Chapters 1–9		

4 Grading Policy

Homework (Eight evenly weighted assignments):	17.5%
Three midterm exams at 27.5% each:	82.5%
Cumulative final exam	27.5%

There will be four exams, equally weighted. The **lowest score** will not be counted. There will be **no make-up exams**. If you miss one exam, you must take the other three. If you miss two or more exams with **valid excuses**, your exam weights will be adjusted accordingly.

Grading will be determined according to both the scale outlined in the RPI policy provided in the [link](#) and the average student performance. Generally, the class average is expected to align with the satisfactory to good range.

Extension or special accommodation requests submitted less than 48 hours before the due date will only



be considered in cases of emergencies such as illness or family situations. The final decision on granting special accommodations rests entirely with the instructor based on the specific circumstances.

5 Absence Policy

Details of the excused absence policy can be found on the Student Success website ([link](#)). If you know you will have an excused absence in advance (funeral, wedding, Rensselaer-sanctioned event, etc.), inform your instructor as early as possible. In the case of illness, no medical excuse is required for absences of up to one week, but you should contact your instructor as soon as possible. Students are responsible for the missed content, and video recordings of lectures will not be provided. In the case of an excused absence on the day of a scheduled midterm exam, students will be eligible for a makeup exam, which will differ from the regular exam, although the content covered will be the same. The timing of makeup exams will be coordinated between the instructors and all affected students. A written make-up exam will not be offered after the cumulative final exam due to time constraints. As needed, the instructor may modify the grading policy or require an oral make-up exam..

6 Academic Integrity

Student-teacher relationships are built on trust. For example, students must trust that teachers made appropriate decisions about the structure and content of the courses they teach, and teachers must trust that the assignments the students turn in are their own. Acts which violate this trust undermine the educational process. The Rensselaer Handbook of Student Rights and Responsibilities defines various forms of Academic Dishonesty and you should make yourself familiar with these. In this class, all assignments that are turned in for grade must represent the student's own work. In cases where help was received, or teamwork was allowed, a notation on the assignment should indicate your collaboration.

Submission of any assignment that is in violation of this policy will result in the penalty of a grade of zero for this assignment. If you have any questions concerning this policy before submitting an assignment, please ask the professor for clarification. If you have any question concerning this policy before submitting an assignment, please ask for clarification.

All students are expected to be familiar with and abide by the policies and procedures, including those on academic integrity, contained in the current version of the Rensselaer Handbook of Student Rights and Responsibilities ([link](#)). Any individual assignments you submit in this course should be entirely your own work. Copying homework or projects will not be tolerated. You are encouraged to first do your assignments by yourself and then discuss solutions with others. Cheating or plagiarism will result in punitive measures. Violations of academic integrity may also be reported to the appropriate Dean (Dean of Students for undergraduate students or the Dean of Graduate Education for graduate students, respectively).

7 Disability Services

Rensselaer Polytechnic Institute strives to make all learning experiences as accessible as possible. If you anticipate or experience academic barriers based on a disability, please let me know immediately so that we can discuss your options. To establish reasonable accommodations, please register with The Office of Disability Services for Students. After registration, make arrangements with the Director of Disability Services as soon as possible to discuss your accommodations so that they may be implemented in a timely fashion. DSS contact information: dss@rpi.edu; +1-518-276-8197; 4226 Academy Hall.



8 Support Services

[PInfo](#) - contains various resource links for students, academic resources, support services, and safety emergency preparedness.

Academic Assistance	ALAC—Advising and Learning Assistance Center	518.276.6269	webpage link
Student Health and Wellness	Counseling Center	518.276.6479	webpage link
	Student Health Center	518.276.6287	webpage link
Student Support Services	Class Deans, Undergraduate Dean, Graduate Experience Dean, Student Success Dean, First Year Experience	518.276.8022	webpage link
DOTCIO (IT Services)	Help Desk, Submit a ticket to IT Services and Support Center	518.276.7777	webpage link